Blog

OT Zero Trust Drives Cyber Safe Vehicles

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In any modern metropolis, having a self-driving car find its own parking space is a dream come true. No more driving around in circles looking for an open spot, finding it, and watching another car zip in front of you and take your hard-won spot. No more dangerously directing your phone app to find a parking spot rather than safely paying

attention to traffic. Your onboard AV valet negotiates all this while you eat out, shop, or enjoy entertainment. Soon, delivery companies will send autonomous vehicles with robots or drones to bring packages to your front, or back, door depending on the quality of their GPS. When that time come, both drivers and fleet owners must feel secure knowing that their vehicles will behave on and off the road. OT zero trust is a lightweight, supercharged solution for automakers.

Within the manufacturing plant, you need systems that can adapt to what's happening at any given moment. Malware cannot be allowed to stop production, or sneak on board either vehicles or the Mothership fleet control center so it can wreak havoc when the rubber meets the road. Cybersecurity protections must make sure that interconnected auto assembly line robots or cobots, that may have been trustworthy yesterday or five minutes ago, can still be trusted right now. OT zero trust keeps track of trusted entities and grants access privileges based on the current situation and your security policies. The four cornerstones of OT zero trust explained below are the powertrain of cybersecurity for traditional autos and AVs.

1: Inspect

The moment a new robot or device arrives for deployment on the assembly line, inspect it using the OT zero trust portable security device. This wipes away any malware and takes an inventory of onboard apps. Automakers and auto parts manufacturers use this portable security device to inspect devices running outdated operating systems such as Windows 2000, 7, or XP. Portable security devices are rugged and can be used even in harsh factory environments. Portable security devices can also make secure file transfers when you need to load data onto air-gapped systems. You can use them to perform routine or surprise security inspections without throttling the performance of your equipment or slowing down the assembly line.

2: Lockdown

Think of trust lists as adaptive stoplights. Stoplights adjust traffic flow based on current driving conditions. In the automotive manufacturing plant, trustworthiness is evaluated based on the current flow of the assembly line. Only trustworthy messages get the green light.

3: Segment

Divide your network with safety “cone zones” and throttle messages through trust lists designed to take the best route based on the current network traffic conditions, the production line situation, and your access control policies. Network segments can be used to sandbox digital twins and protect against corrupt feedback loops. Any traffic that could compromise the system can be quarantined from the network, minimizing the damage incurred.

4: Reinforce

Barricade systems by stopping known and unknown attacks. Even zero day threats are zapped by virtual patching while device manufacturers develop and test security patches. Machine-learning reasons about where new threats are ramping up, based on analysis of the latest trends. With OT zero trust, anti-malware computations travel in the fast lane while malware gets caught in speed traps to make sure that attacks are jammed and production processes always have a green light for safety.

OT zero trust provides cybersecurity protections that add a new dimension to vehicle quality. By following our core philosophy of never trust, always verify, OT zero trust assembles trustworthy auto manufacturing processes that build trustworthy AVs.

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